

```
Processando config 24, seed 3, teste: Rabbit
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Rabbit: nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed3.bin
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 2, n = 8388608, r = 0, s = 32, L = 38,
Sparse = TRUE
```

```
Number of bits = n = 8388608
Number of cells = 2^L = 274877906944
Expected number per cell = 1 / 32768
EColl = n^2 / (2k) = 128
Hashing = TRUE
```

```
Collision test
CollisionOver: density = n / k = 1 / 32768.00
Expected number of collisions = Mu = 128.00
```

```
-----
Results of CollisionOver test:
```

```
POISSON approximation :
Expected number of collisions = N*Mu : 256.00
Observed number of collisions : 276
p-value of test : 0.11
```

```
-----
Total number of cells containing j balls
```

```
j = 0 : 549739036948
j = 1 : 16776664
j = 2 : 276
j = 3 : 0
j = 4 : 0
j = 5 : 0
```

```
-----
CPU time used : 00:00:04.50
```

```
Generator state:
16777216 bits have been read.
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed3.bin
```

snpair_ClosePairsBitMatch test:

N = 1, n = 262144, r = 0, t = 2

Max of all bit distances : 17.00
p-value of test : 0.39

CPU time used : 00:00:00.04

Generator state:
33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed3.bin

snpair_ClosePairsBitMatch test:

N = 1, n = 131072, r = 0, t = 4

Max of all bit distances : 8.00
p-value of test : 0.12

CPU time used : 00:00:00.08

Generator state:
50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed3.bin

svaria_AppearanceSpacings test:

N = 1, Q = 524288, K = 524288, r = 0, s = 30, L = 15

Sequences of $n = (K + Q)L = 15728640$ bits
Q = 524288 initialization blocks
K = 524288 blocks for the test
the blocks have L = 15 bits

Normal statistic : 0.16
p-value of test : 0.44

CPU time used : 00:00:00.00

Generator state:
67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed3.bin

scomp_LinearComp test:

N = 1, n = 18176, r = 0, s = 32

Number of degrees of freedom : 7
Chi2 statistic for size of jumps : 6.38
p-value of test : 0.50

Normal statistic for number of jumps : -52.96
p-value of test : 1 - eps1 *****

CPU time used : 00:00:00.05

Generator state:
67127040 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed3.bin

scomp_LempelZiv test:

N = 1, n = 16777216, r = 0, s = 32, k = 24

Normal statistic : -0.32
p-value of test : 0.62

CPU time used : 00:00:00.41

Generator state:

83904256 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed3.bin

sspectral_Fourier1 test:

N = 1, n = 1048576, r = 0, s = 32, k = 20

Normal statistic : -0.90
p-value of test : 0.82

CPU time used : 00:00:00.20

Generator state:

84952832 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed3.bin

sspectral_Fourier3 test:

N = 8065, n = 2048, r = 0, s = 32, k = 11

Kolmogorov-Smirnov+ statistic = D+ : 0.023
p-value of test : 0.58

Kolmogorov-Smirnov- statistic = D- : 0.025
p-value of test : 0.53

Anderson-Darling statistic = A2 : 0.54
p-value of test : 0.71

CPU time used : 00:00:00.12

Generator state:

101728032 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed3.bin

sstring_LongestHeadRun test:

N = 1, n = 60, r = 0, s = 32, L = 279616

Number of degrees of freedom : 3
Chi-square statistic : 3.14
p-value of test : 0.37

Global longest run of 1 : 27.00
p-value of test : 0.06

CPU time used : 00:00:00.07

Generator state:
118504992 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed3.bin

sstring_PeriodsInStrings test:

N = 1, n = 524288, r = 0, s = 31

Number of degrees of freedom : 94
Chi-square statistic : 92.62
p-value of test : 0.52

CPU time used : 00:00:00.00

Generator state:
135282208 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed3.bin

sstring_HammingWeight test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Number of degrees of freedom : 22
Chi-square statistic : 42.46
p-value of test : 5.5e-3

CPU time used : 00:00:00.01

Generator state:
152059424 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed3.bin

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : 1.33
p-value of test : 0.09

CPU time used : 00:00:00.00

Generator state:
168836640 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed3.bin

sstring_HammingCorr test:

N = 1, n = 262144, r = 0, s = 32, L = 64

Normal statistic : -0.22
p-value of test : 0.59

CPU time used : 00:00:00.00

Generator state:
185613856 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed3.bin

sstring_HammingCorr test:

N = 1, n = 131072, r = 0, s = 32, L = 128

Normal statistic : 1.43
p-value of test : 0.08

CPU time used : 00:00:00.00

Generator state:
202391072 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed3.bin

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 173.40
p-value of test : 0.56

CPU time used : 00:00:00.01

Generator state:
219168288 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed3.bin

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 353.94
p-value of test : 0.03

CPU time used : 00:00:00.00

Generator state:

235945504 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed3.bin

sstring_HammingIndep test:

N = 1, n = 131072, r = 0, s = 32, L = 64, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 489
Chi-square statistic : 521.46
p-value of test : 0.15

CPU time used : 00:00:00.01

Generator state:

252722720 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed3.bin

sstring_AutoCor test:

N = 1, n = 16777153, r = 0, s = 32, d = 1

```
-----
Normal statistic          :   -0.41
p-value of test          :    0.66
-----
```

```
CPU time used            :   00:00:00.00
```

```
Generator state:
 269499904 bits have been read.
```

```
*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:   /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed3.bin
```

```
sstring_AutoCor test:
```

```
-----
      N =  1,   n = 16777154,   r =  0,   s = 32,   d = 2
-----
```

```
-----
Normal statistic          :   -0.99
p-value of test          :    0.84
-----
```

```
CPU time used            :   00:00:00.02
```

```
Generator state:
 286277088 bits have been read.
```

```
*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:   /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed3.bin
```

```
sstring_Run test:
```

```
-----
      N =  1,   n = 3355443,   r =  0,   s =  32
-----
```

```
-----
Total number of 1 runs:  3355451
```

```
Number of degrees of freedom      :   36
Chi2 statistic for number of runs :   32.07
p-value of test                   :    0.66
-----
```

```
-----
Total number of bits:  13419744
```

```
Normal statistic for number of bits :   -0.39
p-value of test                   :    0.65
```

CPU time used : 00:00:00.04

Generator state:
299696832 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed3.bin

smarsa_MatrixRank test:

N = 1, n = 16384, r = 0, s = 32, L = 32, k = 32

Number of degrees of freedom : 3
Chi-square statistic : 0.62
p-value of test : 0.89

CPU time used : 00:00:00.05

Generator state:
316474048 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed3.bin

smarsa_MatrixRank test:

N = 1, n = 163, r = 0, s = 32, L = 320, k = 320

Number of degrees of freedom : 2
Chi-square statistic : 1056.73
p-value of test : eps *****

CPU time used : 00:00:00.03

Generator state:
333165248 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed3.bin

swalk_RandomWalk1 test:

N = 1, n = 131072, r = 0, s = 32, L0 = 128, L1 = 128

Test on the values of the Statistic H

Number of degrees of freedom	:	42
ChiSquare statistic	:	36.43
p-value of test	:	0.71

Test on the values of the Statistic M

Number of degrees of freedom	:	42
ChiSquare statistic	:	26.40
p-value of test	:	0.97

Test on the values of the Statistic J

Number of degrees of freedom	:	64
ChiSquare statistic	:	54.15
p-value of test	:	0.81

Test on the values of the Statistic R

Number of degrees of freedom	:	36
ChiSquare statistic	:	53.84
p-value of test	:	0.03

Test on the values of the Statistic C

Number of degrees of freedom	:	22
ChiSquare statistic	:	17.08
p-value of test	:	0.76

CPU time used : 00:00:00.05

Generator state:
349942464 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed3.bin

swalk_RandomWalk1 test:

N = 1, n = 16384, r = 0, s = 32, L0 = 1024, L1 = 1024

Test on the values of the Statistic H

Number of degrees of freedom	:	94
ChiSquare statistic	:	91.92
p-value of test	:	0.54

Test on the values of the Statistic M

Number of degrees of freedom	:	94
ChiSquare statistic	:	96.79
p-value of test	:	0.40

Test on the values of the Statistic J

Number of degrees of freedom	:	512
ChiSquare statistic	:	531.40
p-value of test	:	0.27

Test on the values of the Statistic R

Number of degrees of freedom	:	90
ChiSquare statistic	:	82.66
p-value of test	:	0.70

Test on the values of the Statistic C

Number of degrees of freedom	:	50
ChiSquare statistic	:	48.89
p-value of test	:	0.52

CPU time used : 00:00:00.03

Generator state:

366719680 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed3.bin

swalk_RandomWalk1 test:

N = 1, n = 1675, r = 0, s = 32, L0 = 10016, L1 = 10016

Test on the values of the Statistic H

Number of degrees of freedom	:	130
ChiSquare statistic	:	117.04
p-value of test	:	0.79

Test on the values of the Statistic M

Number of degrees of freedom	:	131
ChiSquare statistic	:	127.22
p-value of test	:	0.58

Test on the values of the Statistic J

Number of degrees of freedom	:	161
ChiSquare statistic	:	181.31
p-value of test	:	0.13

Test on the values of the Statistic R

Number of degrees of freedom	:	130
ChiSquare statistic	:	128.86
p-value of test	:	0.51

Test on the values of the Statistic C

Number of degrees of freedom	:	89
ChiSquare statistic	:	102.93
p-value of test	:	0.15

CPU time used : 00:00:00.03

Generator state:

383496480 bits have been read.

===== Summary results of Rabbit =====

Version:	TestU01 1.2.3
Generator:	ufile_CreateReadBin
Number of bits:	16777216
Number of statistics:	39

Total CPU time: 00:00:05.98
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

Test	p-value
-----	-----
5 LinearComp	1 - eps1
22 MatrixRank, 320 x 320	eps
-----	-----

All other tests were passed